Meridian — Final Project Report

Apartment Complex Management System

Team 98 · IITM BS Degree, Software Engineering · May 2026 term

Repository: https://github.com/V47VANSH/MAY2026-Team-098

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1. Executive Summary

Meridian is a platform for running an apartment society. It replaces the spreadsheets, WhatsApp groups and paper registers that societies typically use with a single system covering maintenance billing and payment, complaint management, shared-facility booking, visitor and emergency handling, society communication, and remote oversight for owners who do not live on site.

The product decision that shaped everything else was to build it as **multi-tenant SaaS** rather than for one society. One deployment serves many societies, each isolated by `tenant_id`. This was a deliberate expansion beyond the literal Milestone 1 brief, taken because it is a materially stronger engineering problem — tenant isolation, scope derived from token claims, a platform-level admin tier — without meaningfully increasing feature-build time, and because it matches the shape of real PropTech products. The isolation guarantee is enforced by a dedicated test suite that attempts every cross-society access path with a token belonging to a different society, so it holds by test rather than by convention.

What was delivered:

Backend	FastAPI, 9 routers, **48 API paths / 59 operations / 62 schemas**
API contract	OpenAPI 3.1 generated from the live route table, with 24 user stories machine-mapped to endpoints and 44 error codes catalogued
Database	18 tables, 13 indexes, Alembic migrations
Frontend	React + TypeScript SPA, every screen rendering from the real API
ML	Separate FastAPI service: complaint classifier and payment-risk scorer, both trained
Tests	**149 backend + 21 ML service**, all passing (7 further backend tests run only against a live Redis)
CI	Three GitHub Actions workflows: lint, typecheck, test, and contract-drift gates

All six Milestone 1 epics are implemented end to end. The system runs from a clone with no external service, no database server and no credentials: SQLite is the default and seeds a demo society on first start.

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2. Team and Contributions

Contributor	Commits	Primary area
Tushar (`23f2003751`)	40	Backend APIs, ML service, API integration, testing, CI
«name» (`21f3000278`)	7	Frontend — official dashboard, Meridian AI assistant
Vansh Bansal	5	Repository setup, frontend
«name» (`21f1004124`)	4	Frontend pages

> **Before submitting:** replace the two *«name»* placeholders above. Git records only the roll numbers for those accounts, and named ownership is what the teamwork criterion looks for.

56 commits across 9 branches. Ownership was organised horizontally — one person per layer — which the team chose deliberately so that each member worked across all six epics rather than owning one vertical slice. The consequence is that every feature touches several people at once, and the OpenAPI contract is what keeps them unblocked: the backend publishes the contract, the frontend generates its types from it, and CI fails if the two drift apart.

Part A — Requirements and Design

A1. Problem and objective

Apartment societies coordinate maintenance collection, complaints, shared facilities, visitors and emergencies through manual and semi-manual processes. The concrete pain points identified in Milestone 1 user research were: no visibility into what is owed and whether it was received; complaints raised verbally with no trace and no accountability; double-booked facilities; visitors admitted on the strength of a phone call; and owners living elsewhere having no way to see any of it.

The objective is a centralised platform that makes each of these auditable and visible to the people affected.

A2. Users and roles

Nine roles across four tiers. Visitors are deliberately **not** accounts — they hold a signed, time-boxed, single-use pass token, which keeps the auth model simple while still giving a guard something cryptographically verifiable.

Tier	Role	Core capabilities
Primary	`RESIDENT_OWNER`	Pay maintenance, raise complaints, book facilities, issue guest passes, raise SOS
Primary	`SOCIETY_OFFICIAL`	Manage residents, invoices, complaint triage, bookings, announcements
Secondary	`NON_RESIDENT_OWNER`	Track dues and complaints remotely, issue guest passes, read announcements
Secondary	Visitor	Holds a guest pass — not an account
Tertiary	`SECURITY_GUARD`	Verify guest passes, receive and acknowledge SOS
Tertiary	`MAINTENANCE_STAFF` / `CLEANING_STAFF`	Receive and resolve assigned complaints
Tertiary	`SERVICE_PROVIDER`	Fulfil service requests
Platform	`SUPER_ADMIN`	Onboard and manage societies

A3. Epics and user stories

24 user stories across 6 epics. These are not merely listed in a document — they are embedded in the published API contract as an `x-user-story-mapping` extension, so every story names the endpoints that implement it and the mapping travels with the spec.

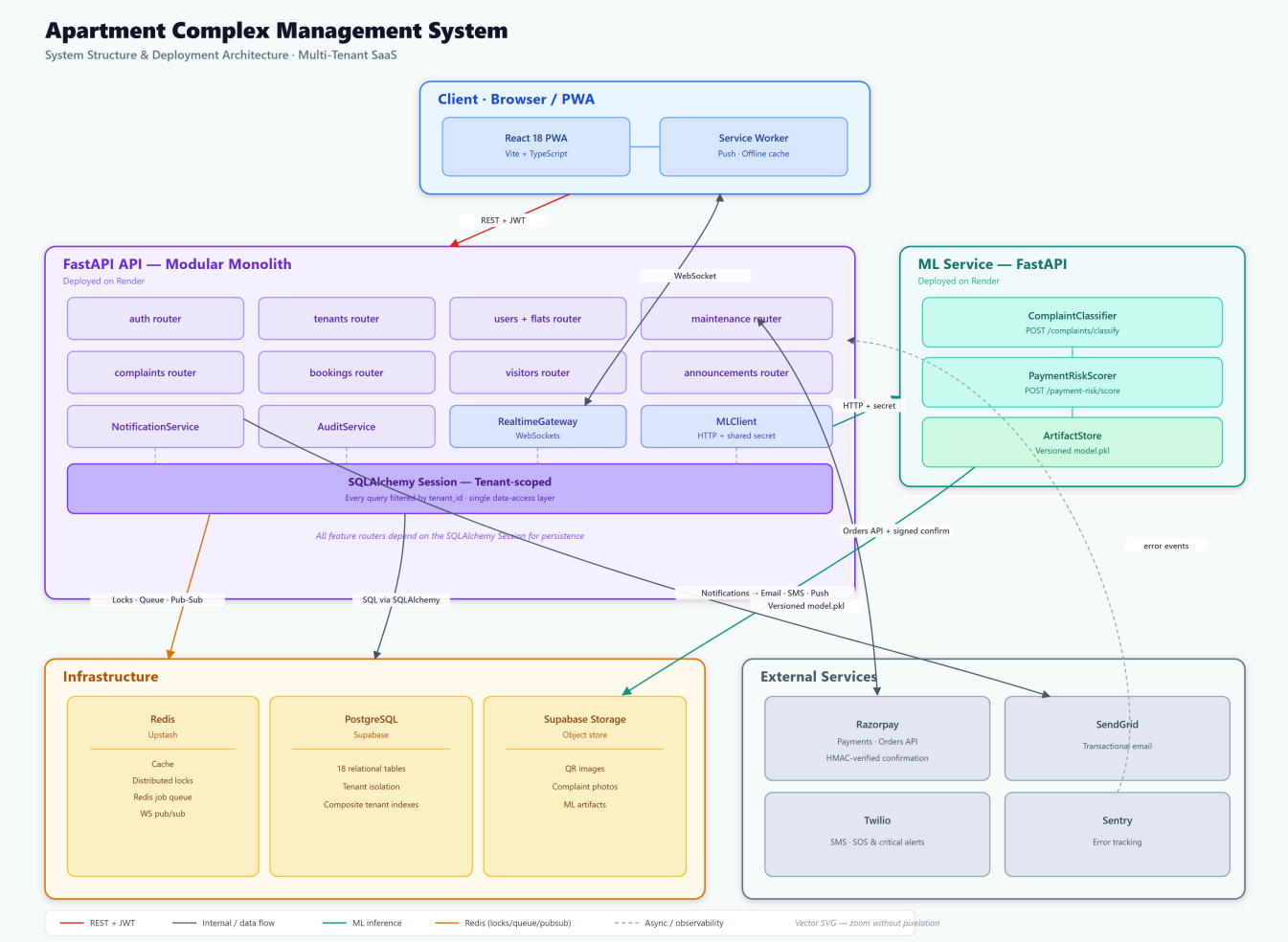
Epic	Stories	Router
E1 — Maintenance Management	4	`maintenance.py`
E2 — Complaint Management	4	`complaints.py`
E3 — Facility Management	4	`bookings.py`
E4 — Visitor & Emergency Management	5	`visitors.py`
E5 — Society Administration	4	`announcements.py`, `tenants.py`
E6 — Non-Resident Owner Services	3	cross-cutting, role-gated

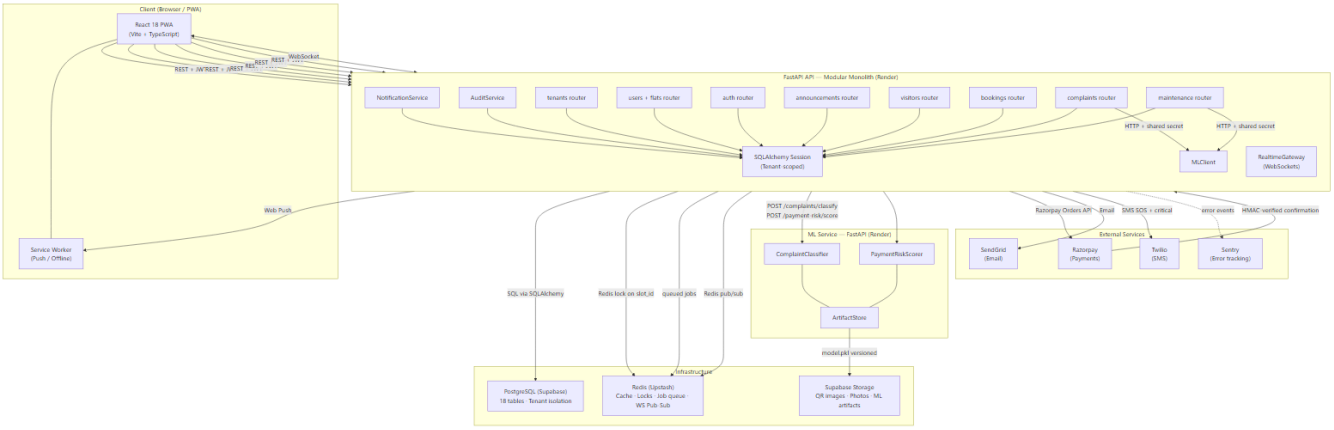
Example, verbatim from `backend/openapi.yaml`:

> **US-1.1** — *As a resident, I want to see my maintenance dues so that I know what I owe and by when.* → `GET /api/v1/maintenance/invoices`, `GET /api/v1/maintenance/invoices/{invoice_id}`, `GET /api/v1/maintenance/payments/{payment_id}`

A4. Architecture

A **modular monolith** for the API — one router per epic — plus a separately deployable ML service. Not microservices: at a five-person scale, one router per epic gives the same separation of concerns without the operational cost of independent deployables. The ML service is the one exception, because one member owns both models end to end and needs to iterate on them without touching API release cycles.



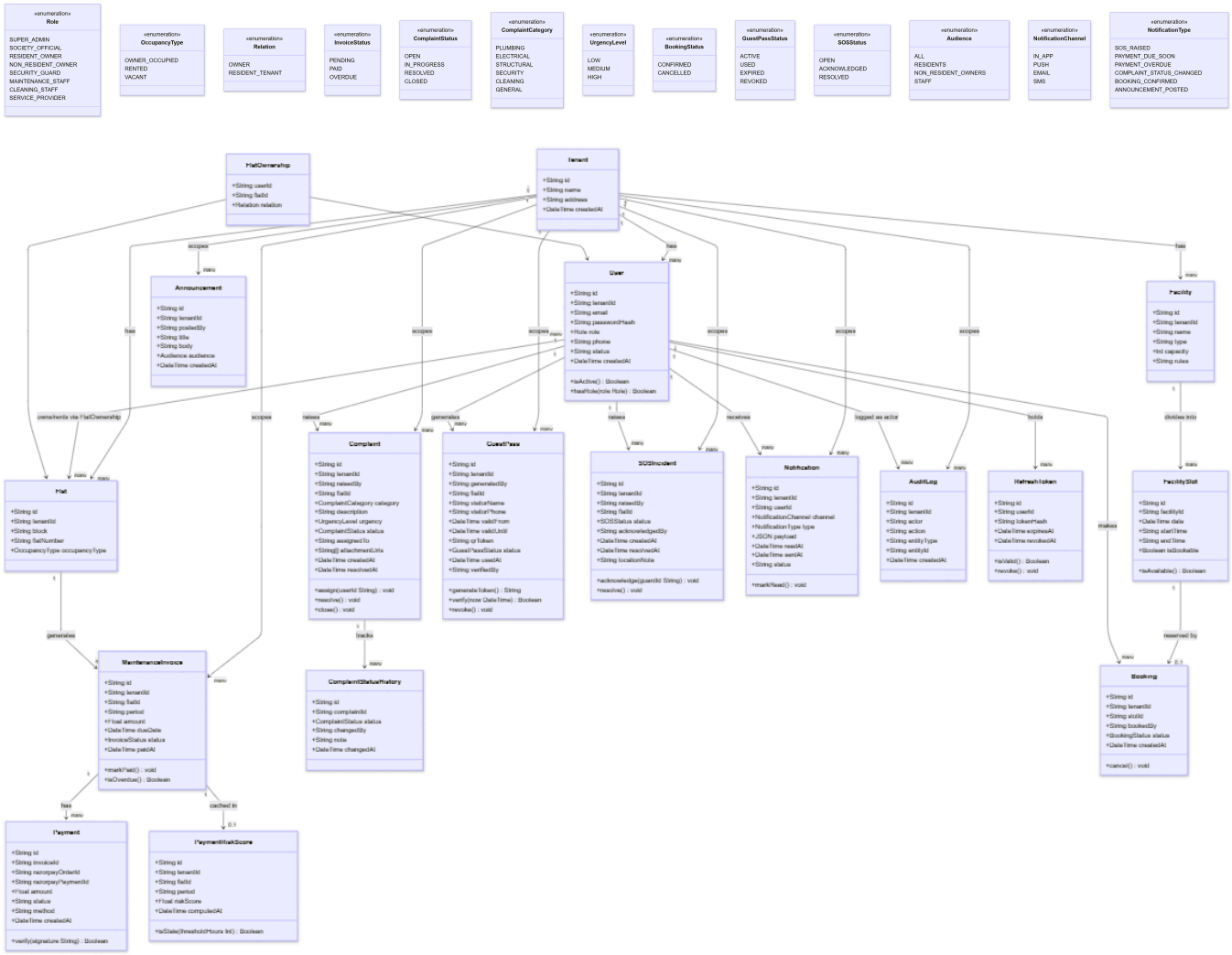


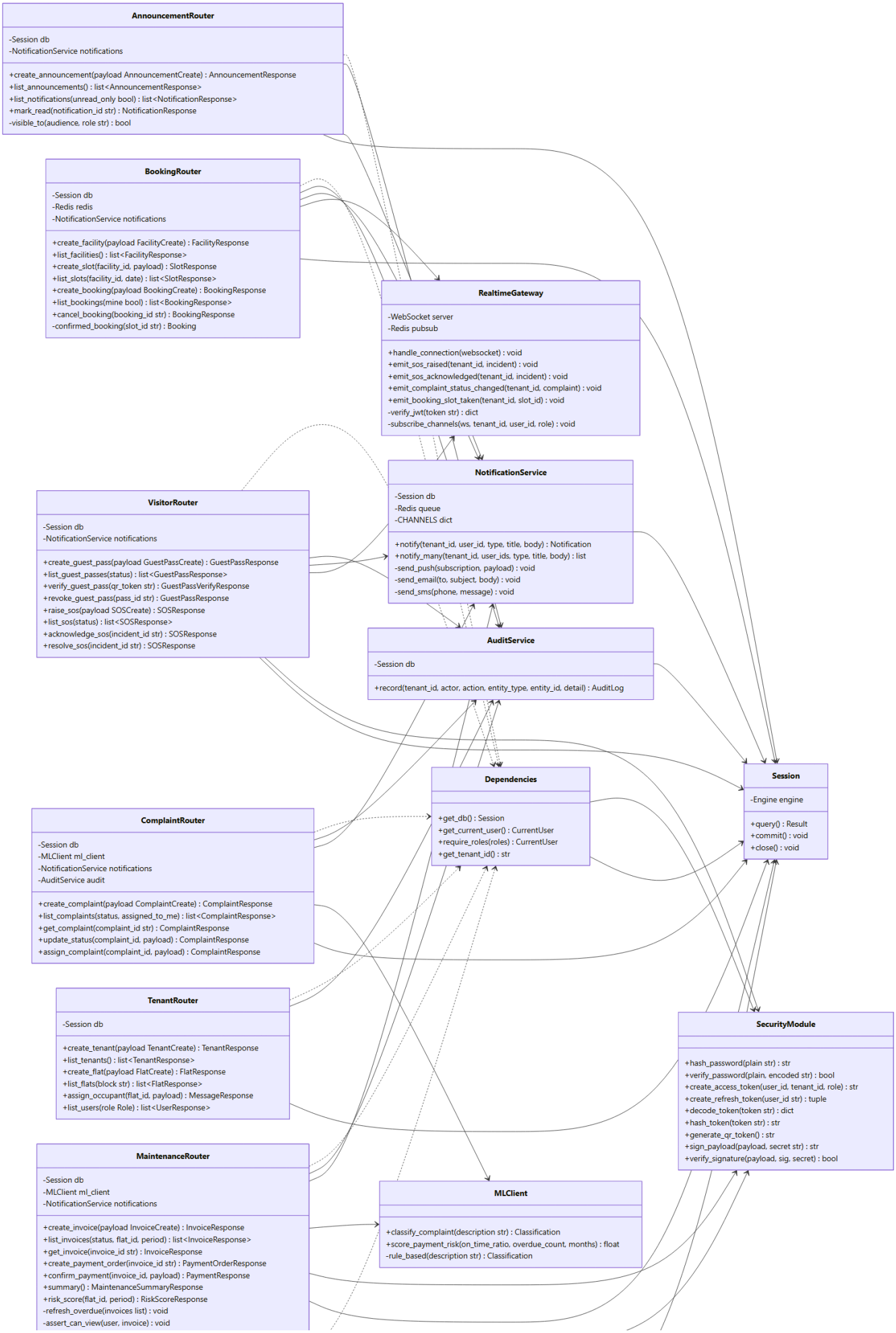
Full detail: 00 — Architecture Overview, 07 — Component Design.

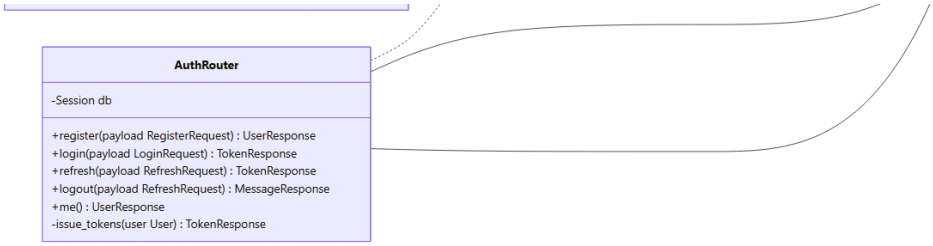
A5. Wireframes and workflow

Screen-by-screen wireframes, the role-based sitemap, and the end-to-end workflow for each epic are in 06 — Wireframes and Workflow.

A6. Class and domain design



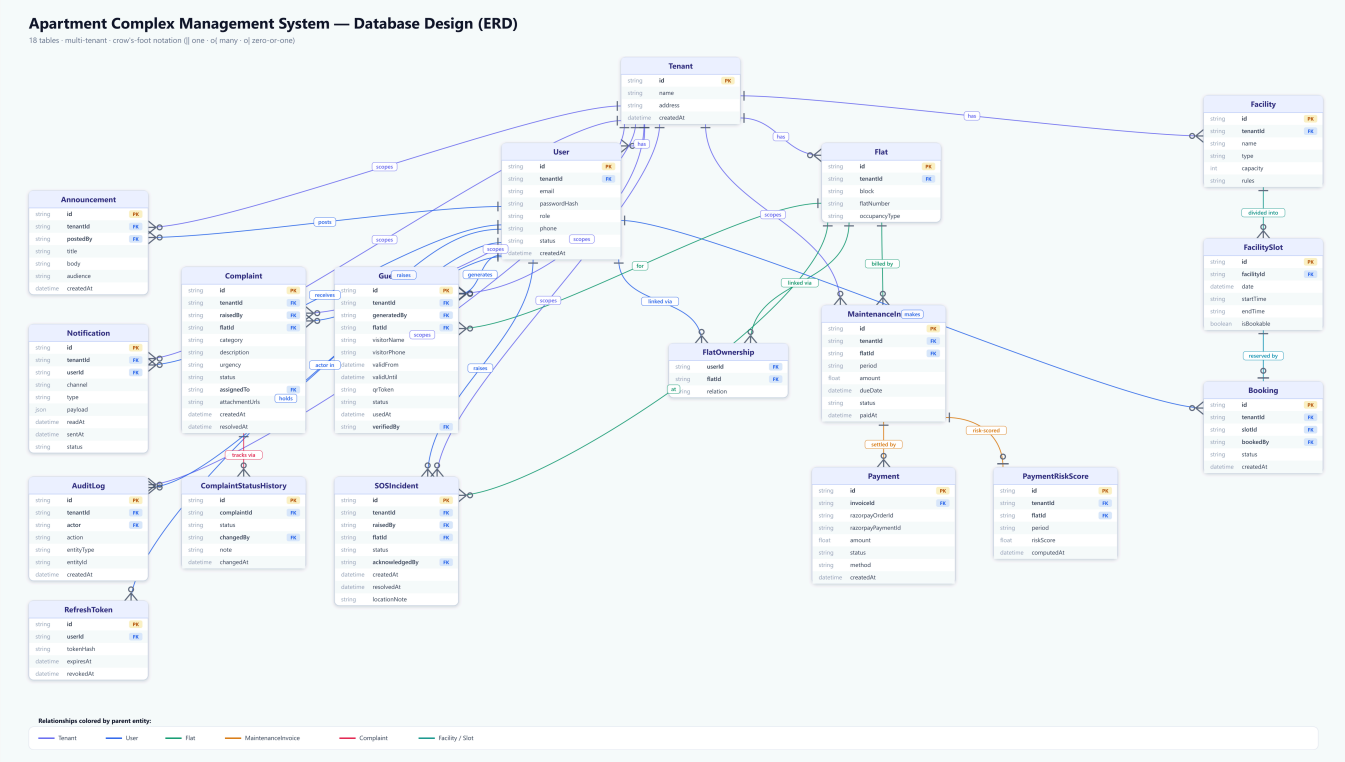

AnnouncementRouter
-Session db
-NotificationService notifications
+create_announcement(payload AnnouncementCreate) : AnnouncementResponse
+list_announcements() : list<AnnouncementResponse>
+list_notifications(unread_only bool) : list<NotificationResponse>
+mark_read(notification_id str) : NotificationResponse
-visible_to(audience, role str) : bool
BookingRouter
-Session db
-Redis redis
-NotificationService notifications
+create_facility(payload FacilityCreate) : FacilityResponse
+list_facilities() : list<FacilityResponse>
+create_slot(facility_id, payload) : SlotResponse
+list_slots(facility_id, date) : list<SlotResponse>
+create_booking(payload BookingCreate) : BookingResponse
+list_bookings(mine bool) : list<BookingResponse>
+cancel_booking(booking_id str) : BookingResponse
-confirmed_booking(slot_id str) : Booking
RealtimeGateway
-WebSocket server
-Redis pubsub
+handle_connection(websocket) : void
+emit_sos_raised(tenant_id, incident) : void
+emit_sos_acknowledged(tenant_id, incident) : void
+emit_complaint_status_changed(tenant_id, complaint) : void
+emit_booking_slot_taken(tenant_id, slot_id) : void
-verify_jwt(token str) : dict
-subscribe_channels(ws, tenant_id, user_id, role) : void
NotificationService
-Session db
-Redis queue
-CHANNELS dict
+notify(tenant_id, user_id, type, title, body) : Notification
+notify_many(tenant_id, user_ids, type, title, body) : list
-send_push(subscription, payload) : void
-send_email(to, subject, body) : void
-send_sms(phone, message) : void
VisitorRouter
-Session db
-NotificationService notifications
+create_guest_pass(payload GuestPassCreate) : GuestPassResponse
+list_guest_passes(status) : list<GuestPassResponse>
+verify_guest_pass(qr_token str) : GuestPassVerifyResponse
+revoke_guest_pass(pass_id str) : GuestPassResponse
+raise_sos(payload SOSCreate) : SOSResponse
+list_sos(status) : list<SOSResponse>
+acknowledge_sos(incident_id str) : SOSResponse
+resolve_sos(incident_id str) : SOSResponse
AuditService
-Session db
+record(tenant_id, actor, action, entity_type, entity_id, detail) : AuditLog
Dependencies
+get_db() : Session
+get_current_user() : CurrentUser
+require_roles(roles) : CurrentUser
+get_tenant_id() : str
Session
-Engine engine
+query() : Result
+commit() : void
+close() : void
ComplaintRouter
-Session db
-MLClient ml_client
-NotificationService notifications
-AuditService audit
+create_complaint(payload ComplaintCreate) : ComplaintResponse
+list_complaints(status, assigned_to_me) : list<ComplaintResponse>
+get_complaint(complaint_id str) : ComplaintResponse
+update_status(complaint_id, payload) : ComplaintResponse
+assign_complaint(complaint_id, payload) : ComplaintResponse
TenantRouter
-Session db
+create_tenant(payload TenantCreate) : TenantResponse
+list_tenants() : list<TenantResponse>
+create_flat(payload FlatCreate) : FlatResponse
+list_flats(block str) : list<FlatResponse>
+assign_occupant(flat_id, payload) : MessageResponse
+list_users(role Role) : list<UserResponse>
SecurityModule
+hash_password(plain str) : str
+verify_password(plain, encoded str) : bool
+create_access_token(user_id, tenant_id, role) : str
+create_refresh_token(user_id str) : tuple
+decode_token(token str) : dict
+hash_token(token str) : str
+generate_qr_token() : str
+sign_payload(payload, secret str) : str
+verify_signature(payload, sig, secret) : bool
MaintenanceRouter
-Session db
-MLClient ml_client
-NotificationService notifications
+create_invoice(payload InvoiceCreate) : InvoiceResponse
+list_invoices(status, flat_id, period) : list<InvoiceResponse>
+get_invoice(invoice_id str) : InvoiceResponse
+create_payment_order(invoice_id str) : PaymentOrderResponse
+confirm_payment(invoice_id, payload) : PaymentResponse
+summary() : MaintenanceSummaryResponse
+risk_score(flat_id, period) : RiskScoreResponse
-refresh_overdue(invoices list) : void
-assert_can_view(user, invoice) : void
MLClient
+classify_complaint(description str) : Classification
+score_payment_risk(on_time_ratio, overdue_count, months) : float
-rule_based(description str) : Classification



Full detail: 08 — Class Diagram.

A7. Database design

18 tables. Every tenant-scoped table carries `tenant_id`, and every composite index leads with it.



Full detail: 09 — Database Design, 03 — Database Architecture.

Part B — Implementation

B1. Frontend

React 18 + TypeScript on Vite. Two state systems with a deliberate split: **Zustand** holds session state (the signed-in user and their tokens), **TanStack Query** holds server state. They are not interchangeable — server state needs caching, refetching and invalidation, session state needs none of that and must be readable outside React by the API client.

The API client (`src/api/client.ts`) is the single place the frontend talks to the backend. It attaches the bearer token, unwraps the uniform error body into a typed `ApiError` , and transparently refreshes an expired access token before replaying the request. That refresh is single-flight: several queries commonly fail with 401 in the same tick, and because the backend revokes a refresh token as it rotates it, letting each one refresh independently would log the user out.

Request and response types are **generated** from the backend's published spec, not written by hand. CI regenerates them and fails if they differ from what is committed.

Full detail: 01 — Frontend Architecture.

B2. Backend

One `api/v1/*.py` router per epic, wired with `APIRouter` and `Depends()` rather than a global container — so a route's authentication, role and tenant requirements are readable directly from its signature.

Full detail: 02 — Backend Architecture.

B3. API contract

FastAPI generates OpenAPI 3.1 from the Pydantic models on every route, served at `/api/docs` and exported to `backend/openapi.yaml` . Because it is generated from the live route table, it cannot describe an endpoint that does not exist. CI enforces the converse by re-exporting and failing on any diff, so a route cannot be added without the contract following it.

Two extensions beyond standard OpenAPI:

Extension	Contents
`x-user-story-mapping`	6 epics, 24 stories, each naming its endpoints
`x-error-catalogue`	44 error codes with status and meaning

B4. Security

Concern	Implementation
Access tokens	JWT HS256, 15-minute expiry
Refresh tokens	**Rotating** — presenting one issues a new pair and revokes the presented one, so a stolen token is usable at most once and its reuse is detectable. Only the SHA-256 hash is stored, so a database leak yields no usable tokens
Passwords	PBKDF2-SHA256, 260,000 iterations, per-user salt, iteration count embedded in the digest so the cost can be raised later without invalidating existing passwords
Authorisation	`require_roles(...)` dependency reading the token's role claim
Tenant scope	`get_tenant_id` reads the token claim **only** — never a body or query parameter, so a caller cannot ask for another society's data
Guest passes	192-bit random token (not sequential, so passes cannot be enumerated), single-use: the `ACTIVE → USED` flip happens in the same transaction that returns the verification result, which is what defeats replay of a captured QR image
Payments	HMAC-SHA256 over `<order_id>|<payment_id>` verified **before** anything in the payload is trusted

Two deliberate choices worth noting: cross-tenant reads return **404, not 403**, because confirming a record exists would leak that another society holds that id; and login returns an identical error for an unknown email and a wrong password, so the response cannot be used to enumerate registered addresses.

Rate limiting. Fixed-window, keyed on client IP, with tighter ceilings on `/auth/*` (20/min) and `/sos` (10/min) than on general traffic (240/min). The reason differs per endpoint: `/auth` is where credential stuffing and email enumeration are attempted, and `/sos` fans out to every guard in the society over push and SMS, so a loop on it is both a cost and an alert-fatigue problem.

A fixed window admits a burst spanning a boundary — up to twice the limit across two adjacent windows. That is a real weakness, accepted because it costs one `INCR` against Redis where a sliding-window log costs a sorted-set write plus a trim, and the threat here is a sustained loop rather than a precisely timed burst.

`X-Forwarded-For` is believed only behind a proxy explicitly marked trusted. A client can set that header themselves, so trusting it unconditionally would let anyone reset their own counter by varying it — the control would look present and do nothing.

B5. Epic flows

Maintenance and payments. `(tenant_id, flat_id, period)` is unique, so re-running billing cannot double-charge. Pay Now creates a gateway order and a `CREATED` payment row; the invoice only flips to `PAID` on a signature-verified confirmation. Overdue status is recomputed **on read**, so an unpaid invoice past its due date reports `OVERDUE` without depending on a nightly job having run — and that transition is also the moment the flat's occupants are notified.

Complaints. Free text goes to the ML classifier; if it is unavailable, slow, or below the confidence threshold, a deterministic keyword classifier answers instead and the complaint is flagged `needs_manual_triage`.
Submission never depends on ML being up. Status transitions are validated against the lifecycle (`OPEN →`

`IN_PROGRESS → RESOLVED → CLOSED`); a complaint cannot jump from `OPEN` straight to `RESOLVED`, because the intermediate state is what makes the audit trail meaningful. Where staff fix something on the spot, a one-tap action issues both transitions rather than short-cutting the lifecycle.

Bookings — race safety. Double-booking was the concrete failure mode named in Milestone 1. It is defended twice: a per-slot lock serialises concurrent requests, and a **partial unique index** on `slot_id WHERE status = 'CONFIRMED'` makes a second confirmed booking impossible at the database level even if the lock were bypassed by another process. The index is partial so that cancelling frees the slot. Tested with eight genuinely simultaneous threaded requests.

Visitors and emergency. Guest pass tokens are returned only to the resident who created the pass — list views omit them, because a list is not where a secret belongs. Validity is evaluated on read on both the verify and list paths. SOS writes the incident row **first**, then notifies, so the alert survives even if every notification channel fails; a guard acknowledging it notifies the raiser back, which is what turns a fire-and-forget button into a two-way exchange.

B6. Error handling

Every failure — application errors, framework 404s, validation failures, and unhandled exceptions — maps to one shape:

```
{ "statusCode": 409, "error": "SLOT_ALREADY_BOOKED",
  "message": "This slot has already been booked", "requestId": "..." }
```

`error` is a stable machine-readable code the frontend branches on; `message` is prose that may be reworded without breaking clients. `requestId` is generated per request, returned in `X-Request-Id`, and included in the error body — so a user-reported bug traces to its exact server-side log entry. The frontend surfaces it on error screens. The catch-all handler never leaks a stack trace.

B7. Database and migrations

Enums are stored as `VARCHAR` with a `CHECK` constraint rather than native Postgres enum types, which keeps values readable in the database and avoids the migration cost of altering an enum.

Schema has two paths, deliberately: local runs and the test suite build it straight from model metadata so the project clones and runs with no database server, while Postgres is migrated with `alembic upgrade head`. Two paths can drift, so `tests/test_migrations.py` builds both and compares them structurally — columns, foreign keys, indexes and CHECK constraints. A model change committed without a matching revision fails there rather than on deploy.

B8. ML service

Two models, one deployable, called by the backend over HTTP with a shared-secret header.

Complaint classifier — TF-IDF over word unigrams and bigrams into logistic regression, one model per output (category and urgency). Bigrams earn their place: "*not urgent*" and "*no water*" both invert what their individual words suggest. Trained on 107 hand-written seed phrases expanded by openers, locations and urgency cues into ~2,650 labelled examples.

Payment risk — gradient-boosted trees over five features (`past_late_count`, `avg_days_late`, `current_dues_amount`, `tenure_months`, `month`), computed by the backend from invoice history and passed in the request body; the ML service is stateless and never queries the database.

Model	Metric	Floor
Complaint category	macro-F1 **1.0**	0.85
Complaint urgency	macro-F1 **0.9907**	0.60
Payment risk	ROC-AUC **0.6684**	0.65

These numbers need reading carefully, and we would rather say so than let them be misread.

The classifier's macro-F1 of 1.0 is not a generalisation estimate. Its training and test data are generated from the same phrase templates, so that figure largely measures the templates. The meaningful test uses phrasings written nowhere in the training set: the model initially got 4 of 6, and — importantly — returned confidence 0.30 and 0.39 on the two it missed, **below the backend's 0.55 threshold**, so the backend discarded them and flagged the complaints for manual triage. Broadening the seed vocabulary fixed those two. The test now asserts the property that actually matters: *never confidently wrong*, rather than never wrong. A wrong-but-confident answer is the only genuinely damaging outcome, because it routes a complaint to the wrong trade without flagging it.

The risk model's ROC-AUC of 0.6684 looks poor until measured against the **Bayes-optimal 0.6818** for its own generative process — it reaches 98% of what is achievable. Outcomes are sampled from a probability, so most of the remaining gap is irreducible noise rather than a modelling failure. The trainer records both numbers so the headline is not read out of context. An earlier floor of 0.75 was unreachable by construction and was corrected to 0.65.

Both models are trained on **bootstrapped and simulated data**. Retraining on real complaint text and real invoice history is what would turn them into evidence.

Full detail: 04 — ML Architecture.

B9. GenAI integration — the Meridian AI assistant

`POST /api/v1/ai/assistant` accepts natural language and returns a reply plus a structured intent. Backed by **Google Gemini** when `GEMINI_API_KEY` is set, and by rule-based intent matching when it is not — either way it classifies the request.

The assistant's job is working out what someone wants; the screens already do the work. It maps the returned intent to the route that handles it, per role, and hands the user there. An intent a role has no screen for produces no link rather than one that bounces off a route guard.

(This feature post-dates the Milestone 2 design documents and appears in none of them; it is documented here.)

B10. Dashboard aggregation

Three role-scoped endpoints (`/dashboard/resident` , `/official` , `/non-resident`) each replace the four or five calls a dashboard would otherwise fan out, with the server doing the joining — so the counts on a dashboard cannot drift from the lists they summarise. *(Also added after Milestone 2 and documented here.)*

Part C — Testing and Quality

C1. Strategy

`pytest` with FastAPI's `TestClient`, one suite per router. **Each test runs against its own throwaway SQLite database**, so tests share no state and run in any order — no PostgreSQL, no Redis, no running server. This is what makes the suite runnable by a grader with a fresh clone and nothing installed.

C2. Current results

```
$ cd backend && pytest -q
149 passed, 7 skipped

$ cd ml-service && pytest -q
21 passed

$ cd backend && ruff check .
All checks passed!

$ cd frontend && npm run typecheck
(clean)
```

Backend coverage by module: `test_ai` 7, `test_announcements` 10, `test_auth` 14, `test_bookings` 13, `test_complaints` 17, `test_maintenance` 19, `test_migrations` 5, `test_missing_apis` 9, `test_rate_limit` 9, `test_redis_paths` 9, `test_tenant_isolation` 9, `test_user_management` 9, `test_visitors` 19.

The 7 skipped are `test_redis_integration`, which exercises the same guarantees against a **live** Redis rather than a fake. They skip when no Redis is configured — which keeps the suite runnable with nothing installed — and run in CI against a `redis:7` service container. The distinction matters: a fake proves the code issues the commands it intends to, but only a real server can prove that `SET NX PX` is genuinely atomic and that the release script really is compare-and-delete.

C3. Tests that check properties, not endpoints

Three suites are worth reading before the rest, because they assert guarantees rather than responses:

Suite	Guarantee
`test_tenant_isolation.py`	Every cross-society access path, attempted with a token from a *different* society — the realistic shape of the attack
`test_bookings.py`	Eight simultaneous threaded booking requests for one slot produce exactly one `201` and seven `409`s
`test_migrations.py`	The Alembic revisions and the ORM models build structurally identical schemas

C4. Functional test cases

Ten fully documented cases — API, inputs, expected output, actual output, pass/fail — are in 10 — Milestone 3 Report §4, covering registration, wrong-password login, single-use refresh tokens, forged payment signatures, duplicate invoices, complaint auto-classification, illegal status transitions, the concurrent-booking race, guest-pass reuse, and cross-society reads.

C5. The defect our testing found — TC-11

Testing surfaced a genuine defect, which we documented rather than quietly fixed: **an expired guest pass was still listed as `ACTIVE`.** The validity window was evaluated on the verify path but not on the list path, so a pass whose window had closed stayed listed as active until someone scanned it. There was no security impact — the gate refused it either way — but the resident's list was misleading.

It was recorded with root cause and impact in 10 §5, held under a `strict=True` `xfail` marker so that fixing it without removing the marker would fail the build, scheduled as Sprint 5 Priority 1, and **resolved** by applying the same read-time evaluation on both paths. The marker is gone and the case is now a passing positive assertion.

That full loop — found by testing → documented with root cause → tracked → scheduled → fixed → verified — is the single clearest piece of evidence of the team's quality process, which is why the original record is preserved unedited.

Part D — Tools and Technologies

D1. Frontend

Tool	Version	Purpose
React	18.3.1	UI framework
React DOM	18.3.1	DOM renderer
TypeScript	5.5.4	Static typing across the whole app
Vite	5.3.5	Dev server and production bundler
`@vitejs/plugin-react`	4.3.1	React fast-refresh
React Router DOM	6.26.0	Routing and role-based route guards
Zustand	4.5.4	Session state, readable outside React by the API client
TanStack React Query	5.101.4	Server state, caching, invalidation on write
Tailwind CSS	3.4.7	Styling
PostCSS / Autoprefixer	8.4.40 / 10.4.19	CSS pipeline
openapi-typescript	7.13.0	**Generates request/response types from the backend spec**
qrcode	1.5.4	Scannable guest-pass QR rendering

D2. Backend

Tool	Version	Purpose
Python	3.11+ (verified 3.13.1)	Language
FastAPI	0.115.6	Web framework; generates the OpenAPI contract
Uvicorn	0.34.0	ASGI server
SQLAlchemy	2.0.36	Typed ORM
Alembic	1.14.0	Versioned schema migrations
Pydantic	2.10.4	Request/response validation
pydantic-settings	2.7.0	Environment-driven configuration
PyJWT	2.10.1	HS256 access and refresh tokens
passlib	1.7.4	PBKDF2-SHA256 password hashing
email-validator	2.2.0	Email field validation
python-multipart	0.0.20	Form parsing
PyYAML	6.0.2	OpenAPI export
httpx	0.28.1	HTTP client

D3. ML service

Tool	Version	Purpose
scikit-learn	1.6.0	TF-IDF + LogisticRegression; GradientBoostingClassifier
joblib	1.4.2	Model artifact persistence (`model.pkl`)
FastAPI / Uvicorn / Pydantic	as above	Service and contract

D4. Data

Tool	Purpose
SQLite	Zero-configuration default; seeds a demo society on first run
PostgreSQL	Production path via `DATABASE_URL` ; migrations exercised in CI

D5. External APIs

Service	Mode	Usage
Razorpay	Test / sandbox	Order creation and HMAC-SHA256 signature verification before an invoice is settled
Google Gemini	`gemini-2.5-flash`	Natural-language assistant, with rule-based fallback

D6. Testing and quality

Tool	Purpose
pytest 8.3.4	149 backend + 21 ML tests
FastAPI `TestClient`	In-process integration testing
Python `threading`	Genuine concurrency test for the booking race
Ruff	Backend and ML lint gate
`tsc --noEmit`	Frontend type gate

D7. API contract and documentation

OpenAPI 3.1 · Swagger UI (`/api/docs`) · ReDoc (`/api/redoc`) · `scripts/export_openapi.py` · `x-user-story-mapping` · `x-error-catalogue`

D8. DevOps and collaboration

Tool	Purpose
Git / GitHub	Version control, pull requests, code review
GitHub Actions	`ci-backend.yml` , `ci-frontend.yml` , `ci-ml.yml`
Jira	Issue tracking (SCRUM board)
`.env.example`	Committed configuration templates for backend and frontend

D9. Design and modelling

ERD (`db-design.svg`), class diagram, domain-entity diagram, service-class and service-relationship diagrams, system-level component diagram, architecture structure diagram, and an **interactive diagram viewer** (`docs/architecture-diagram-viewer.html`).

D10. Why these choices

Python across backend and ML. The ML service had to be Python. Making the API Python too means one language across both services rather than two, and makes the backend directly testable with `pytest` — the tool the milestone deliverables are graded with.

FastAPI over Flask. The OpenAPI contract is generated from the route definitions themselves, which is what makes contract-first collaboration between the frontend and backend owners possible without a hand-maintained spec drifting.

SQLite by default. The submission must run for a grader with no database server. PostgreSQL remains one environment variable away and is exercised in CI.

Multi-tenant from the start. Retrofitting tenant isolation is far harder than building with it.

Part E — Code Review, Issue Reporting and Tracking

E1. Process

```
Jira issue (SCRUM-xx)
   → feature branch
      → commits referencing the issue id
         → pull request
            → CI gates: ruff · pytest · tsc · contract-drift checks
               → peer review and approval
                  → merge to main
```

`main` is the always-runnable branch. Work happens on feature branches and reaches `main` only through a reviewed pull request that CI has passed.

E2. Tools

Jira for issue tracking (SCRUM board, sprint planning, burndown). **GitHub** for code review via pull requests. **GitHub Actions** as the automated first-pass reviewer — lint, tests, typecheck and contract-drift run before a human looks at the diff.

E3. Traceability — Jira issue to commit

Thirteen commits carry their Jira issue id in the subject line, giving a direct trace from a board item to the code that implemented it:

Jira	Commit	Date	Author	Subject
SCRUM-18	`084129f`	2026-07-14	21f3000278	Improved society official dashboard and UX improvements
SCRUM-24	`5c16820`	2026-08-02	Tushar	Identify APIs from user stories
SCRUM-25	`9b36016`	2026-08-02	Tushar	Define endpoints, inputs, and outputs
SCRUM-25	`30a448d`	2026-08-02	Tushar	Align design docs and diagrams with the implemented FastAPI stack
SCRUM-26	`99434df`	2026-08-02	Tushar	Create Swagger-compatible YAML documentation
SCRUM-27	`177d230`	2026-08-02	Tushar	Implement backend API endpoints
SCRUM-28	`978bc65`	2026-08-02	Tushar	Add validation and error handling
SCRUM-29	`eee99fe`	2026-08-02	Tushar	Integrate external / GenAI APIs
SCRUM-30	`4ab8684`	2026-08-02	Tushar	Design functional API test cases
SCRUM-31	`fac1bfb`	2026-08-02	Tushar	Compare expected vs actual outputs
SCRUM-32	`735d6a0`	2026-08-02	Tushar	Implement pytest unit tests
SCRUM-33	`ed06ed2`	2026-08-02	Tushar	Demonstrate features to users and collect feedback
SCRUM-46	`0f2f30f`	2026-07-14	21f3000278	Implement Meridian AI

E4. Branch inventory

Branch	Purpose
`main`	Always-runnable trunk
`UI-V1` , `UI-V2`	Two iterations of the interface
`sprint4`	Sprint 4 backend work
`feature/api-integration`	Frontend wired to the live API
`feature/Dashboard-improvements`	Official dashboard rework
`feature/Meridian-AI`	GenAI assistant
`feature/Additional-API`	Sprint 5 backend and integration
`docs/final-submission-alignment`	Final submission documentation

E5. Project chronology

Date	Activity
5–14 Jul	Repository setup, UI iterations, official dashboard, Meridian AI
2 Aug	**Sprint 4** — SCRUM-24 → 33: API design, implementation, YAML, test cases, pytest, user feedback
9–12 Aug	Dashboard improvements, API integration groundwork
21–23 Aug	**Sprint 5** — full frontend/API integration, ML service, Alembic, CI, doc alignment

E6. Defect lifecycle — a worked example

The TC-11 guest-pass defect (Part C5) is the process working end to end:

Stage	Evidence
Found	User-testing feedback item 7 and the API test suite
Reported	Documented with root cause, impact and reproduction in `docs/10` §5
Tracked	Held under a `strict=True` `xfail` so it could not be silently fixed
Planned	Scheduled as Sprint 5 Priority 1 in `docs/10` §7
Fixed	Read-time window evaluation applied on the list path
Verified	Marker removed; case now passes as a positive assertion; suite green

E7. User feedback and its effect

Seven feedback items were collected by demonstrating the working endpoints to users (10 §6). Four confirmed existing behaviour. One was **declined with a reasoned alternative**: an official asked to skip `IN_PROGRESS` when staff fix something on the spot, and the team kept the intermediate state because it is what makes the audit trail meaningful, delivering a one-tap UI action that issues both transitions instead. One became a new filter (`?needs_triage=true`). One was confirmed as the TC-11 defect above. All three accepted items were delivered in Sprint 5.

E8. Screenshots

> *Insert here:* Jira board and backlog · sprint list with dates · burndown or velocity chart · 2–3 issue detail pages showing the Activity/History tab (SCRUM-31 and SCRUM-32 are the most illustrative) · sprint report · GitHub pull requests with their review threads · branch list · Actions tab showing green runs · Insights/Contributors graph.

Part F — Instructions to Run the Application

F1. Prerequisites

- **Node.js 18+** and npm
- **Python 3.11+**

Verified on Python 3.13.1, Node 22.12.0, npm 11.7.0.

No database server, message broker or cloud account is required.

F2. Backend (start this first)

Windows

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
uvicorn app.main:app --reload
```

Ubuntu / macOS

```
cd backend
python3 -m venv .venv
source .venv/bin/activate
pip3 install -r requirements.txt
uvicorn app.main:app --reload
```

The API starts on http://localhost:8000. On first run it creates `backend/acms.db` and seeds a demo society — nothing else to install.

URL	What it is
http://localhost:8000/api/docs	Interactive Swagger UI
http://localhost:8000/api/redoc	ReDoc reference
http://localhost:8000/api/openapi.json	Machine-readable spec
http://localhost:8000/health	Liveness probe

F3. Frontend

In a second terminal:

```
cd frontend
npm install
npm run dev
```

Open http://localhost:5173. Requests to `/api` are proxied to the backend, so the browser only ever talks to one origin and no CORS configuration is needed.

F4. ML service (optional)

Skip it and the application still works — complaint classification falls back to the keyword classifier and flags each complaint for manual triage, which is the designed behaviour rather than a failure.

```
cd ml-service
python -m venv .venv
.venv\Scripts\activate          # Windows
source .venv/bin/activate       # Ubuntu / macOS
pip install -r requirements.txt
uvicorn app.main:app --port 8100
```

Trained artifacts are committed, so it serves predictions immediately. To retrain:

```
python -m app.complaints.train
python -m app.payment_risk.train
```

Then point the backend at it and restart the backend:

```
set ML_SERVICE_URL=http://localhost:8100          :: Windows
export ML_SERVICE_URL=http://localhost:8100       # Ubuntu / macOS
```

F5. Demo accounts

Every seeded account uses the password **`Password123`**. The sign-in page lists them behind **Show demo accounts** and fills the form when one is clicked.

Email	Role
`resident@greenvalley.example`	Resident / Flat Owner (flat B-304)
`official@greenvalley.example`	Society Official
`owner@greenvalley.example`	Non-Resident Owner
`guard@greenvalley.example`	Security Guard
`staff@greenvalley.example`	Maintenance / Cleaning Staff
`admin@platform.example`	Platform admin (API only)

F6. Running the tests

```
cd backend      && pytest        # 149 tests
cd ml-service   && pytest        # 21 tests
cd frontend     && npm run typecheck && npm run build
```

F7. Verification of these instructions

These steps were not written from memory. They were verified on 23 Aug 2026 by cloning `main` into an empty directory and following this section top to bottom on fresh virtual environments:

Step	Result
`git clone` → repository contents	`backend/`, `frontend/`, `ml-service/`, `docs/`, `.github/` present; no `.venv`, `node_modules` or database file
Backend: venv → `pip install -r requirements.txt` → `pytest`	**149 passed**, 7 skipped
Backend: `uvicorn app.main:app`	`/health` → `{"status":"ok"}`; `/api/docs` → 200; demo login → 200
ML service: venv → install → `pytest`	**21 passed**
Frontend: `npm install` → `npm run typecheck`	Clean
Frontend: `npm run build`	Built in 23.6 s
Frontend: `npm run dev` → sign-in through the API proxy	200

One thing to expect: the very first request through the Vite proxy immediately after `npm run dev` can time out while the dev server finishes starting. A refresh resolves it — it is a cold start, not a failure.

F8. Two configuration caveats

Online card payment needs real Razorpay credentials. The backend verifies the gateway's HMAC before settling an invoice — correctly, since otherwise any client could mark an invoice paid for free. With the placeholder key the order is still recorded and the resident is told online payment cannot complete, rather than being shown a false receipt. The society official's **offline payment** path works fully and is the one to use in a demo.

The AI assistant uses rule-based intent matching unless `GEMINI_API_KEY` is set. Either way it classifies the request and links to the right screen.

Part G — Design Evolution and Deviations

Where the delivered system differs from the Milestone 2 design, each difference is a decision. The pattern in all of them is the same: **the feature degrades to a working fallback rather than a missing capability.**

Designed	Delivered	Why
Redis distributed lock for bookings	**Delivered**, with an in-process fallback	Redis is used when `REDIS_URL` is set and falls back otherwise. The fallback is not a lesser version of the same thing: an in-process lock serialises within one worker and does nothing across several, so what Redis buys is correctness across processes. Either way the partial unique index remains the actual guarantee
Rate limiting	**Delivered**	Fixed-window, per client IP, tighter on `/auth/*` and `/sos`. See B4 for the fixed-versus-sliding tradeoff and why `X-Forwarded-For` is only trusted behind a declared proxy
WebSocket gateway for live updates	Polling	Every REST endpoint behind the intended events exists and is tested. The transport is the only missing piece, and polling is correct if less elegant
Queued push/email/SMS dispatch	Channel routing implemented, logged at the dispatch boundary	Real delivery needs provider credentials. The in-app notification row is always written, so no notification is lost — only its external copy
Supabase Storage for attachments	Not built	Complaint photo upload is not implemented because there is no object store to put a file in. `attachment_urls` exists on the model but no endpoint writes to it
Entity methods on models (doc 08)	Behaviour in the router/service layer	The class diagram documents the conceptual model; the implementation keeps domain logic where the transaction boundary is. A deliberate anaemic-model choice, not an oversight
`Notification.payload` (JSON)	`title` + `body` columns	Every notification the system sends is a title and a body. A JSON column would have been unused generality
PWA and Sentry	Not built	Scope traded for depth in the delivered features. Listed in Part I

Two of these moved during the final sprint. Rate limiting and the Redis-backed lock were both documented as designed and were not implemented; both now are. The Redis work is worth a note on how it was verified, because the tests found two defects the implementation would otherwise have shipped with:

- Releasing the lock used Redis's `EVAL` unconditionally. Some managed Redis offerings disable scripting; the release then failed and the lock sat until its TTL expired, serialising every booking for that slot behind it. It now falls back to a read-then-delete, accepting a rare mistaken release because the database index is the real guarantee and a guaranteed stall is far worse.
- More seriously, a *contended* lock and an *unreachable* Redis were handled identically — both fell through to the in-process lock — so a request that lost the race barged into the critical section anyway. Two threads were measurably inside it at once. Contention now waits; unavailability falls back.

Neither would have been caught by testing against a fake alone, which is the argument for the live-Redis suite described in C2.

Part H — Hosting

The system is deployed to the topology 05 — DevOps Architecture specifies:

Component	Host	Notes
Frontend	**Vercel**	Root directory `frontend`, Vite preset
API	**Render**	Root directory `backend`, migrations run before the server starts
ML service	**Render**	Root directory `ml-service`; trained artifacts are committed, so there is no training step in the build
Database	**Supabase Postgres**	Connected through a pooler, not the direct endpoint
Redis	**Upstash**	Optional; backs the booking lock and rate limiter

`render.yaml` and `frontend/vercel.json` are committed, and `DEPLOYMENT.md` is a standalone walkthrough. Everything below is a decision the deployment forced, and each is the kind of thing that fails quietly rather than loudly.

The frontend proxies rather than calling the API directly. Vercel rewrites `/api/*` to Render, so the browser stays same-origin in production exactly as it does behind the Vite dev proxy locally. CORS is therefore never exercised in either environment, and no API URL is compiled into the JavaScript bundle — the same build artifact works against any backend.

Schema ownership had to be made explicit. Startup called `create_all` unconditionally, which is right for SQLite and wrong for a database Alembic manages: it would create tables with no revision stamp, and the next `upgrade` would fail on tables that already exist. `AUTO_CREATE_TABLES` now defaults to true for SQLite and false otherwise, so local runs stay zero-setup and migrations own everything else.

Prepared statements are disabled against Postgres. Supabase's transaction pooler hands out a backend per transaction, so a statement psycopg prepared on one may not exist on the next. That surfaces as intermittent errors, under load, in production only, and never locally against SQLite — so it is turned off, which costs a little planning time per query and makes either pooler mode correct.

Free-tier instances sleep. Render free services spin down after roughly fifteen minutes idle, and the next request takes 30–50 seconds while the container wakes. Worth stating plainly so it is not mistaken for a performance defect during evaluation.

Alongside the hosting, three CI workflows gate every pull request on lint, tests, typecheck, migration integrity and contract drift.

Part I — Limitations and Future Work

Stated plainly, in the order we would tackle them. Nothing here is hidden elsewhere in the report — an undisclosed gap is worth less than a disclosed one, because a reader who finds one has to wonder what else there is.

1. **Frontend test suite.** There is none. `tsc --noEmit` is a real gate, but it is a type gate, not a behaviour gate — it cannot tell you a button stopped calling the right endpoint. Vitest and React Testing Library first, then Playwright for the six golden paths. This is the largest genuine hole in the quality story and the reason it sits at the top.
2. **Real notification delivery.** Channel routing is implemented and logged at the dispatch boundary, but nothing is sent. Wiring a push, email and SMS provider behind the existing `notify()` call, with a queue so a slow provider cannot affect a request.
3. **WebSocket gateway.** Replacing polling for SOS alerts and live booking-slot availability. Every REST endpoint behind the intended events already exists and is tested; only the transport is missing.
4. **ML on real data.** Both models are trained on bootstrapped and simulated data, and the reported metrics should be read as described in B8 rather than at face value. Retraining on accumulated complaint text and invoice history is what would turn them into evidence.
5. **Complaint photo upload.** `attachment_urls` exists on the model, but there is no object store behind it and no endpoint that writes to it. Needs Supabase Storage.
6. **PWA.** Installability, an offline app shell, and Web Push.
7. **Deployment hardening.** Migrations currently run in the start command; Render's dedicated pre-deploy step is the better home for them on a paid plan. Sentry for error tracking, and an uptime monitor.
8. **Smaller gaps.** An audit-log UI (the endpoint exists and nothing renders it), a camera QR scanner for guards (typing the token works today), and hold-to-trigger SOS to prevent accidental presses.

Two items that appeared on this list in earlier drafts — **rate limiting** and the **Redis-backed booking lock** — were both closed in the final sprint and are described in B4 and Part G.

Appendix — Document index

Document	Contents
00 — Architecture Overview	System overview, roles, epic mapping, technology status
01 — Frontend Architecture	React app structure, state, API client
02 — Backend Architecture	Routers, contract, security, per-epic flows, errors, testing
03 — Database Architecture	Tenant isolation, migrations, enums, seeding
04 — ML Architecture	Both models, training, evaluation, fallbacks
05 — DevOps Architecture	CI, environments, secrets, monitoring, hosting status
06 — Wireframes and Workflow	Screens, sitemap, end-to-end flows
07 — Component Design	Component inventory and interactions
08 — Class Diagram	Domain and service class design
09 — Database Design	18-table ERD, keys, constraints, indexes
10 — Milestone 3 Report	Sprint 4 record: API inventory, TC-01…TC-11, feedback, Sprint 5 plan
11 — Final Report	This document
13 — Presentation	Slide deck with speaker notes
14 — Demo Script	Running order for the recorded demonstration